The Effects of Different Surface Designs of Airfoils on Lift and Drag Performance

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ABSTRACT

Airfoils are the cross section of an airplane wing and their various designs come with a wide variety of uses and this exploration into the designs can offer more optimal options for different flight regimes. The material and smoothness of an airfoil are known to determine the lift, drag, and overall efficiency of a plane wing, and surface designs on the airfoil can create different effects on those forces. With the usage of 3D modeling, scaled-down models of airfoils can be produced which then can be used to make different surface designs and analyze its effects on lift and drag. The laminar airflow airfoils design of NASA NLF1015 is a good base design as laminar airflow airfoils are made to minimize overall drag by minimizing skin friction and parasitic drag. Though laminar airflow airfoils slightly increase lift, it is insignificant. Thus, this airfoil will be tested for a baseline performance, then analyzing the specific surface designs: shark skin like design, a wavy surface, triangular prisms along the leading to the trailing edge, and dimples that resemble golf balls. The plain NASA NLF1015 airfoil had the highest performance, confirming its effectiveness in reducing drag through laminar airflow. The second version Triangular prism design is second both statistically significant differences from each other. The wavy and hole textures were similar with them not being significantly different from each other but being significantly different from the others. The bumpy and initial triangular prism are significantly different as bumpy is more efficient than the initial triangular prism design thus making the first triangular prism design the least efficient with the lowest lift-to-drag ratio.

TABLE OF CONTENTS

INTRODUCTION

Overview of Research Problem

The material and smoothness of an airfoil are known to determine the lift, drag, and overall efficiency of a plane wing, and surface designs on the airfoil can create different effects on those forces. With the usage of 3D modeling, scaled-down models of airfoils can be produced which then can be used to make different surface designs and analyze its effects on lift and drag. The laminar airflow airfoils design of NASA NLF1015 (Figure 1) is a good base design as laminar airflow airfoils are made to minimize overall drag by as much as 70% (Bowers, 2011) by minimizing skin friction and parasitic drag. Though laminar airflow airfoils slightly increase lift, it is insignificant. Thus, this airfoil will be tested for a baseline performance, then analyzing the different surface areas and their effects on lift and drag can be easily recognized to then make statistically significant claims about the effects of different designs.

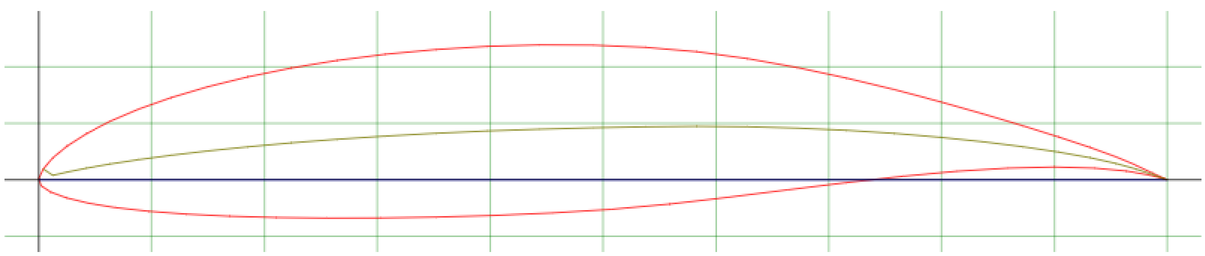

Figure 1 This is a photo of the NASA NLF 1015 airfoil and its geometry. ***NASA NLF1015 (nlf1015-il)*****. (2024). Airfoiltools.com. http://airfoiltools.com/airfoil/details?airfoil=nlf1015-il**

Review of Literature

Lift and drag are fundamental forces that make flight possible for airplanes. Lift is an outcome of a lower pressure above the airfoil and a higher pressure below. Drag is caused by anything that opposes the plane's motion or airflow. Those drag and lift forces can be influenced by factors like surface texture, smoothness, and the general shape of the airfoil (Ujjin, Ngaongam, Chaikaindee, Boonyanetra, 2019) and this difference can be seen after the printing of the airfoil (Figure 2). Surface texture can help delay the flow separation which would lead to more lift, less drag, and ultimately a more efficient design. The surface texture also helps control airflow leading to stability as well. Analyzing the relationship between surface textures and aerodynamic efficiency in airfoils can lead to optimization in drag reduction and lift efficiency.

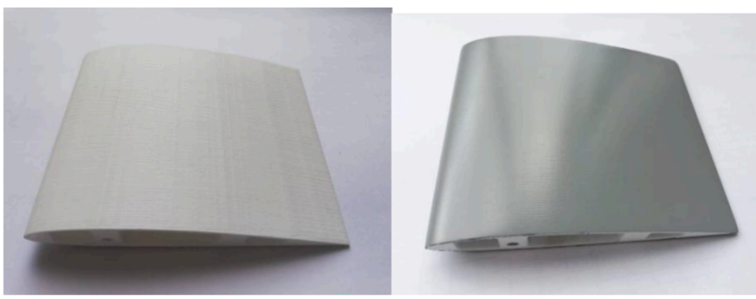

Figure 2 One of their airfoils before (left) and after (right) the acrylic paint finish was applied)

There are different types of airflows when testing or analyzing airfoil performance: laminar flow and turbulent flow. Laminar flow is usually where airflow is usually smooth and regular as it moves smoothly along a streamline. Turbulent flow is where the streamlines break up causing irregular and random airflow (Anderson, 2016). There are airfoils like NASA's NLF1015 (Figure 1) which is a laminar flow airfoil meaning it minimizes drag though it may be difficult to maintain at higher speeds or angles of attack. An angle of attack is the angle at which the chord, an imaginary line connecting the leading and trailing edge of an airfoil, of the airfoil meets the air coming. Turbulent flow airfoils may cause more drag but provide better lift control under turbulent conditions. Thus selecting the right airfoil is crucial to accurately analyze the effects of surface textures.

An airfoil's edges can be key in determining how smoothly air flows across its surface thus influencing lift and drag later on in the airfoil. Previous studies also on leading-edge designs indicate that the shape and efficiency can affect initial airflow attachment to the airfoil (Laitone, 1997). If designed poorly, the trailing edge causes the airflow to spin into whirlpools called vertices which leads to turbulent air behind the airfoil. When this happens, the swirling air pulls energy from the main airflow thus making it harder for the airfoil to move smoothly, causing more drag and lower efficiency. By modifying the edges, it is possible to control the lift-to-drag ratio as the different designs interact with the air differently so depending on the situation you're designing for, you could choose a more optimal lift-to-drag ratio showing the importance of edge designs.

The material or coating used in an airfoil can affect its interaction with the air leading to an impact on both drag and lift. For example, hydrophobic coatings reduce drag by repelling moisture which can be mainly beneficial in humid environments or higher altitudes where moisture piling up on the wing can lead to drag. Other possible coatings can also influence the airfoil as they ultimately help with better efficiency and service life (Goward, 1998). Durability and flexibility in materials can determine how airflow adheres to the surface as it can influence flow separation. Different materials can be used for heat-resistant properties, wear-resistant properties, or more efficient at higher speeds. These all can affect drag in each of their respective ways showing that material and coating choices are to be considered when optimizing airfoil performance.

Wind tunnel testing is a very common method of testing airfoils that allows researchers to analyze airplanes as if they were moving to measure forces like lift and drag. 3D printing allows easy and quick prototyping and modifying of the airfoils to change surface textures or other variables for production and testing in the wind tunnel to analyze it (Szwedziak, Łusiak, Bąbel, Winiarski, Podsędek, Doležal, Niedbała, 2022). In Szwedziak's experiment a model M-346 aircraft was 3D printed for testing (Figure 3) and they measured its lift and drag with varying angles to test the suitability of 3D printing for future tests. They found that lift and drag measurements aligned with theoretical measurements thus validating 3D printing for testing (Figure 4) Wind tunnel testing of these 3D-printed airfoils provides data to then be able to make theoretical predictions about the effects of different surface designs and textures. Thus, the integration of 3D printing airfoils and being able to easily modify and test in wind tunnels for controlled environments is key in gathering data and predicting measurable and practical outcomes.

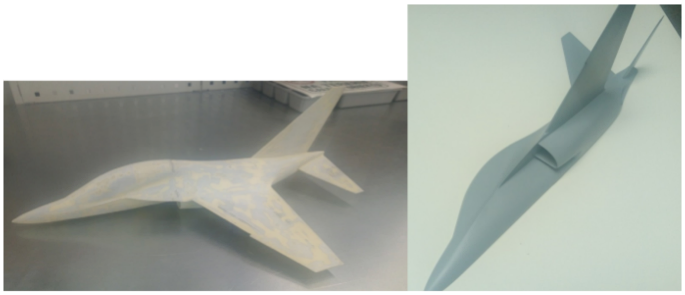

Figure 3 shows the M-346 aircraft model when printed

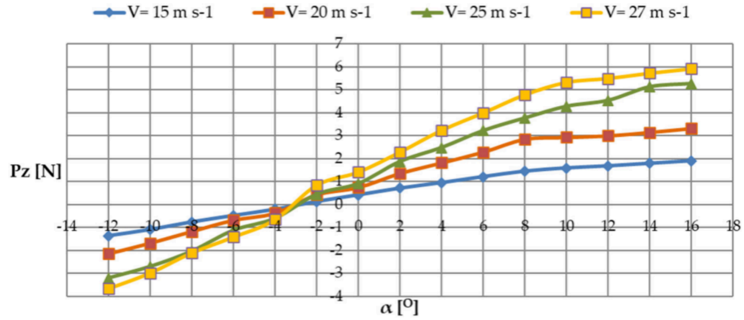

Figure 4 displays the plot of the lift force for all tested positive and negative angles of attack

Understanding and being able to predict the effects of surface designs on lift and drag has been used for the aviation industry, UAV technology, and more. Being able to create designs to customize lift-to-drag ratios based on the needs of the aircraft can help promote fuel efficiency and job effectiveness. These reductions in fuel consumption while increasing operational range take into account sustainability as there is a push towards sustainability in current markets. For UAV applications, different surface designs can vary maneuverability and stability as needed for each mission (Belobaba, Cooper, Seabridge, 2017). With the continued study in this field, more intuitive designs may emerge as they focus on flight conditions, versatility, efficiency, and more.

This experiment utilized these ideas to build off of and begin designing new airfoils to be tested. An initial version of airfoils with varying surface designs were created. These initial designs included a shark skin like design, a wavy surface, triangular prisms along the leading to the trailing edge, and dimples that resemble golf balls. These were three-dimensionally designed and printed. And because they were three-dimensionally designed, they were easy to modify and print to have consistent test subjects (Szwedziak, Łusiak, Bąbel, Winiarski, Podsędek, Doležal, Niedbała, 2022). They were then placed in a wind tunnel with force sensors hooked up to them. When the wind tunnel was turned on, force readings appeared and after some repetition in trials, lift-to-drag ratios for each design were estimated. The designs with the highest ratios moved on and were modified to aim to make them even higher. This gave us a broad generalization of how surface designs of an airfoil influenced its lift-to-drag ratio and this is the basic premise of the experiment.

Problem Statement

- Despite research into the effects of materials and surface roughness on aerodynamic performance, there is a lack of research to do with specific surface designs, particularly dealing with improving the airfoils aerodynamic performance by producing a higher lift-to-drag ratio.

Objective

- This project has created specific surface designs (shark skin like design, a wavy surface, triangular prisms on top of the airfoil, and dimples that resemble golf balls) and analyzed the airfoils lift-to-drag ratio to delve deeper into the designs with higher ratios to introduce research in this field of airfoils.

Hypothesis

- Designs like wavy surfaces and shark skin surface designs will cause higher drag and thus lower lift-to-drag ratios as they extrude out and cause less laminar airflow due to their designs blocking air whereas golf ball like dimples and triangular prisms may reduce drag and increase the lift-to-drag ratio as they allow air to pass around the surface more smoothly.

METHODOLOGY

My Role

Over the course of the 15 weeks, I was responsible for all testing and producing of airfoils. I was also responsible for analyzing the data into cohesive conclusions about the different airfoil designs. The Olathe North Distinguished Scholars Academy supplied my research with the resources to 3D print and test the designed airfoils.

3D Modeling Process

The airfoils were based off of NASA NLF1015 as it is a good base design due to its abilities to minimize skin friction and parasitic drag. In this experiment, four initial designs of surface designs to put on this base airfoil: shark skin/bumpy like design (Figure 5), wavy surface (Figure 6), triangular prisms on the top (Figure 7), and dimples that resemble golf balls (on display). These designs were created by using Chat GPT to write the python coding to input into FreeCAD's software. The airfoils dimensions were kept constant where the chord of the airfoil was 152 mm and the width was 73 mm as it is a reasonable size to be able to create forces of lift and drag to then analyze. These dimensions also apply to the second version designs that were created based off of the initial designs performance. The second version design was based off of the initial triangular prism as it had the worst performance out of all designs and the new design had the triangular prisms run along the leading to the trailing edge (Figure 8).

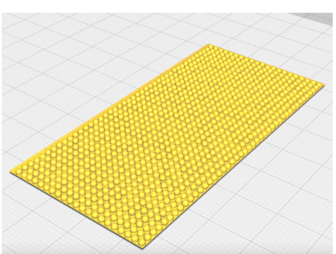

Figure 5 shark skin/bumpy like design

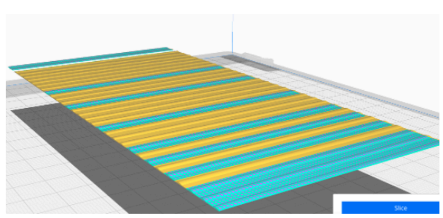

Figure 6 Wavy surface

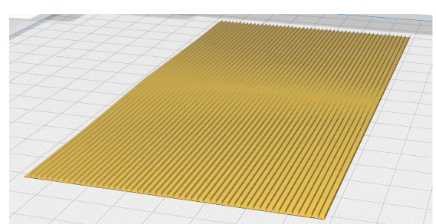

Figure 7 triangular prisms on the top

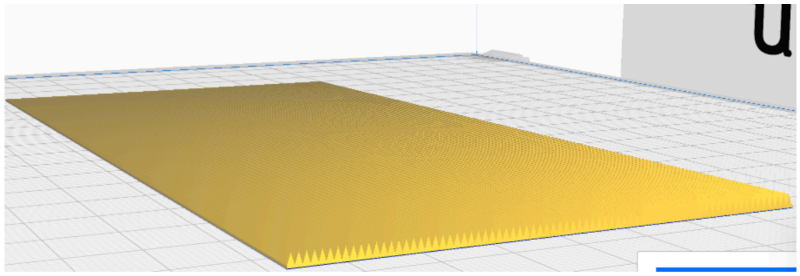

Figure 8 Second Version Triangular Prism

A hole was made at a 10-degree angle on the underside of the airfoil so that the force sensor could fit in it tightly and be hooked up. They were made before the printed skins were glued onto the base airfoils by heating up a metal rod and marking the angle it needs to go in. The skins were then glued on with Gorilla Glue Super Glue Micro Precise and let rest until dry for testing to begin.

<u>3D Printing and Wind Tunnel Testing</u>

The UltiMaker S3 was used to print the airfoils using white PLA filament to the specified dimensions and designs. There was no sanding or smoothening out of the airfoils for consistency. These printed airfoils were then put into a wind tunnel consisting of a Lasko fan on its highest speed through a closed-circuit wind tunnel made of cardboard (Figure 9). The force sensor was clamped on a metal rod going through a box to be up vertically going into the underside of the airfoil so that when the lift force pushes the airfoil up, the force sensor records the difference in the force to give us the lift coefficient. Another force sensor was clamped onto metal rods so that it was positioned behind the airfoil horizontally to measure drag (Figure 10). These measurements were recorded over ten trials and for thirty seconds each and the force sensors were plugged in at the same time so I could get readings of both forces together.

Figure 9 The set up with the fan and tunnel

Figure 10 This shows the placement of the airfoil.

Materials

- FreeCAD software
- ChatGPT
- White PLA filament
- UltiMaker S3 3D printer
- Wind tunnel comprised of cardboard box and Lasko fan
- 2 Vernier Force Sensors with rubber end
- Box and metal rods to align the force sensor
- Vernier Graphical Analysis
- Gorilla Glue

RESULTS

3D Modeling Process

Using ChatGPT for the wavy surface, I asked it “now do one with a wavy surface with a base of being 6 in by 3 inches and the whole thing being height under 1 mm” and after some messing with the wavelength, frequency, orientation and more with errors in the code when inputted I got a code like Figure 11 where I pasted it into FreeCAD and got my skin. I did this with all the skins and produced and sliced the products for printing (Figures 5-8).

🔽 **FreeCAD Python Script (Corrugated Wave Design)**

python   Copy   Edit

```python
import FreeCAD as App
import Part
import numpy as np

# Convert inches to mm
inch_to_mm = 25.4
width = 6 * inch_to_mm  # 6 inches in mm
height = 3 * inch_to_mm  # 3 inches in mm
base_thickness = 0.2  # Flat base thickness (in mm)
max_total_thickness = 1  # Maximum total thickness (including waves)
wave_amplitude = (max_total_thickness - base_thickness) / 2  # Peak wave
wave_length = 5  # Shorter wavelength for distinct wave pattern
num_waves = int(width / wave_length)  # Number of waves in the width

# Create base plate
plate = Part.makeBox(width, height, base_thickness)

# Create a 2D wave profile for extrusion
profile = []
num_points = 50  # More points for smoothness
x_values = np.linspace(0, width, num_points)

for x in x_values:
    z = base_thickness + wave_amplitude * np.sin(2 * np.pi * num_waves *
    profile.append(App.Vector(x, 0, z))

# Convert points into a BSpline curve
wave_curve = Part.BSplineCurve()
wave_curve.buildFromPoles(profile)

# Extrude wave profile along the depth direction
wave_face = wave_curve.toShape().extrude(App.Vector(0, height, 0))

# Fuse with base plate
final_plate = plate.fuse(wave_face)

# Show the final object in FreeCAD
Part.show(final_plate)

App.ActiveDocument.recompute()
```

Figure 11 This is the code given to me that I copied and pasted into FreeCAD for the skins to be rendered

After trial and error and tampering with the settings, skins were produced (Figure 12) where it printed big enough to have small details but small enough to not be as rigid as it is flexible and can wrap onto the airfoil.

Figure 12 This shows the fitment of he skins onto the airfoils as they bend to the curvature of the airfoil

3D Printing and Wind Tunnel Testing

When set to run for thirty seconds, graphical analysis collected data as I left the fan going and waited for both sensors to collect data. I repeated this ten times for each lift and drag for every foil. I then took the average lift and drag for each and then also found the standard deviation for both forces. I compiled this into a table (Figure 13) and then found the average lift-to-drag ratio by dividing the lift value by the drag value. For the error bars I used Google Sheets to find my standard deviation and used the error propagation formula to take into account both standard deviations and have my error bars where p=0.005 and I take into account 2 standard deviations to make the graph shown in Figure 14.

		Wavy	
	Drag		Lift
	0.31	1	0.38
	0.3	2	0.38
	0.29	3	0.36
	0.27	4	0.37
	0.31	5	0.35
	0.3	6	0.42
	0.3	7	0.39
	0.29	8	0.36
	0.32	9	0.38
	0.27	10	0.39
Average	0.296		0.378
Average Lift to Drag Ratio		1.27702703	
Standard Dev:	0.01646545		0.01988858
		ERROR BARS =0.025819888976	

		Holes	
	Drag		Lift
	0.25	1	0.32
	0.25	2	0.33
	0.27	3	0.33
	0.23	4	0.33
	0.25	5	0.34
	0.25	6	0.29
	0.33	7	0.31
	0.26	8	0.32
	0.25	9	0.32
	0.25	10	0.33
Average	0.259		0.322
Average Lift to Drag Ratio		1.243243243	
Standard Dev:	0.026853512		0.013984118
		ERROR BARS 0.030276503542	

		Base	
	Drag		Lift
	0.18	1	0.46
	0.19	2	0.5
	0.22	3	0.51
	0.19	4	0.46
	0.19	5	0.47
	0.19	6	0.47
	0.2	7	0.49
	0.21	8	0.51
	0.24	9	0.49
	0.19	10	0.49
Average	0.2		0.485
Average Lift to Drag Ratio		2.425	
Standard Dev:	0.018257419		0.019002924
		ERROR BARS 0.02635231383	

		Bumpy	
	Drag		Lift
	0.35	1	0.29
	0.36	2	0.3
	0.39	3	0.28
	0.38	4	0.3
	0.35	5	0.31
	0.35	6	0.3
	0.36	7	0.32
	0.36	8	0.32
	0.37	9	0.3
	0.37	10	0.31
Average	0.364		0.303
Average Lift to Drag Ratio		0.832417582	
Standard Dev:	0.013498971		0.012516656
		ERROR BARS 0.018408935025	

		Triangular Pris 1	
	Drag		Lift
	0.45	1	0.33
	0.45	2	0.31
	0.45	3	0.31
	0.44	4	0.32
	0.41	5	0.33
	0.45	6	0.34
	0.46	7	0.33
	0.45	8	0.33
	0.45	9	0.33
	0.44	10	0.33
Average	0.445		0.326
Average Lift to Drag Ratio		0.73258427	
Standard Dev:	0.013540064		0.009660918
		ERROR BARS 0.016633299935	

		Tri Pris 2	
	Drag		Lift
	0.21	1	0.34
	0.23	2	0.33
	0.22	3	0.35
	0.22	4	0.32
	0.21	5	0.33
	0.17	6	0.36
	0.2	7	0.33
	0.2	8	0.35
	0.21	9	0.34
	0.23	10	0.34
Average	0.21		0.339
Average Lift to Drag Ratio		1.614285714	
Standard Dev:	0.017638342		0.01197219
		ERROR BARS 0.021317702605	

Figure 13 This shows the data tables compiled for the different textures as they all follow the same instructions and format.

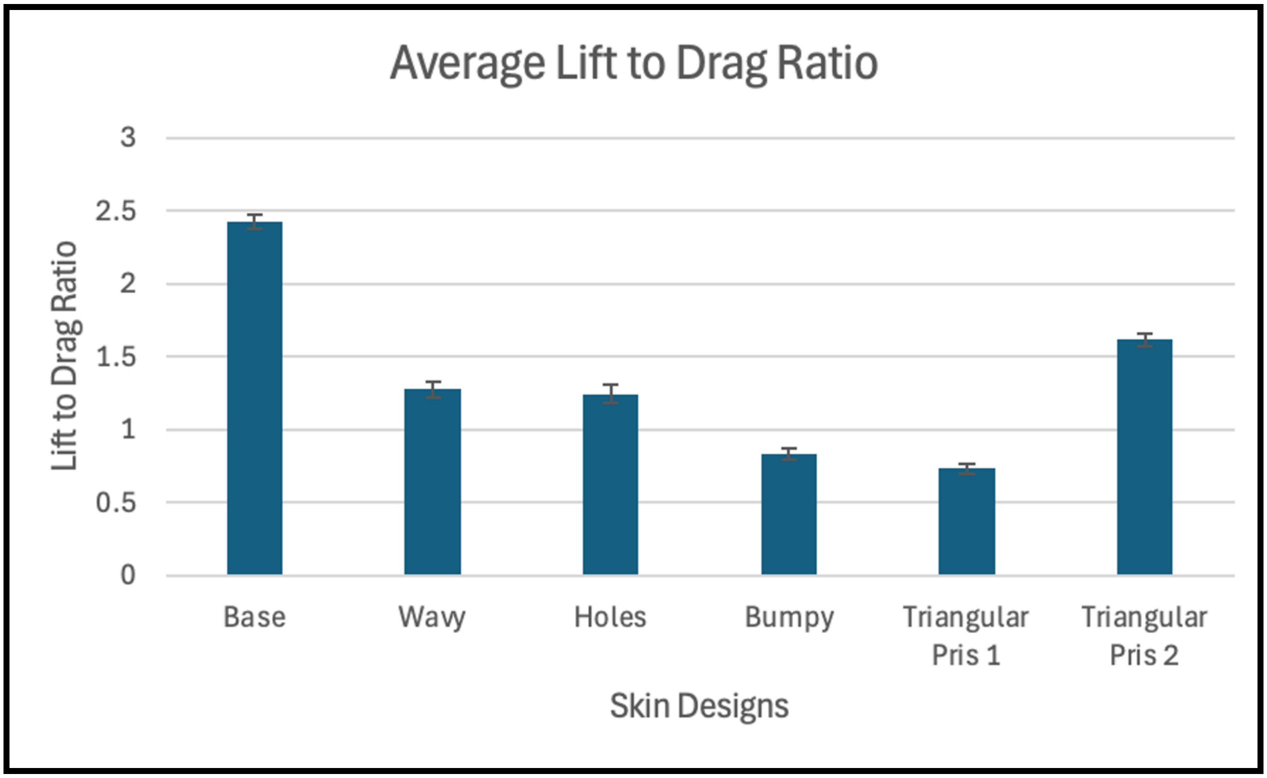

Figure 14 This is the graph for aerodynamic performance based on efficiency with the error bars for each of the designs.

DISCUSSION

Analysis of Data

Interpretation of Data

The data clearly demonstrates how different airfoil surface designs affect lift-to-drag ratios. The plain NASA NLF1015 airfoil had the highest performance, confirming its effectiveness in reducing drag through laminar airflow. The second version Triangular prism design is second both statistically significant differences from each other. The wavy and hole textures were similar with them not being significantly different from each other but being significantly different from the others. The bumpy and initial triangular prism are significantly different as bumpy is more efficient than the initial triangular prism design thus making the first triangular prism design the least efficient with the lowest lift-to-drag ratio.

Objectives and Hypothesis

The objective of testing specific surface designs (shark-skin, wavy surface, triangular prisms, and dimples) was met.

Hypothesis evaluation:

- Increased drag and lower performance for wavy and shark-skin designs were confirmed.
- Golf-ball-like dimples did not significantly enhance performance, refuting this hypothesis.
- Triangular prisms initially reduced efficiency but improved after realignment, partially supporting the hypothesis.

Statistical Significance

Lift-to-drag ratios, standard deviations, and error bars (95% confidence interval) confirmed statistically significant differences between the designs. This straightforward statistical method suited the controlled experimental setup.

Limitations

- 3D printing challenges came up like including gluing the skin onto the airfoil but most designs turned out to be flexible with some needing some force but still fitting well.

- Maintaining a laminar air flow or keeping it constant but using a hard and smooth constructed tunnel with the cardboard fixed this as I got consistent and reliable data.

- Maintaining the 10-degree angle of attack but making the hole at a 10-degree angle helped secure it at the right angle.
- Designing the skins to be enough to render and print was done by simplifying designs to not crash the software.

Past Studies

Similar to Ujjin et al. (2019), this study focused on 3D printing airfoils and using wind tunnel testing to evaluate reliability, validating the effectiveness of 3D printing methods in testing. Unlike Ujjin et al., who primarily focused on the accuracy of printed airfoil geometry, this study significantly specifically explored the aerodynamic efficiency of distinct surface textures by their lift-to-drag ratios.

Implications and Applications

These results can be applied to improve airfoil designs as it can reduce fuel costs and emissions significantly. Understanding how these designs affect the aerodynamic properties can help pilots better control their planes as they adapt their speeds to parts of the flight. Airlines could benefit by reduced operational costs, a smaller carbon footprint, and potentially lowering fare costs for customers.

Future Research

Future research should explore adaptive surface materials to dynamically control lift-to-drag ratios, test real-world fuel consumption and weather impacts to delve more into efficiency for more long-term plans, assess maintenance requirements of these airfoils, and examine noise-reduction airfoils to improve airport conditions.

CONCLUSION

This study aimed to 3D print and test specific airfoil surface designs (shark-skin, wavy surfaces, triangular prisms, and golf-ball-like dimples) to measure their impact on lift-to-drag ratios. Utilizing 3D modeling and wind tunnel testing, the study used the NASA NLF1015 airfoil as a base airfoil. The base and smooth airfoil was the most efficient with the initial triangular prism design being the least efficient. These findings advance understanding of aerodynamic surface effects showing the importance for aircraft efficiency, safety, and environmental sustainability. This research ultimately contributes valuable contributions toward achieving more efficient, cost-effective, and environmentally friendly flying.

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